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Reactivity of ethanol over hydroxyapatite-based Ca-enriched catalysts with various carbonate contents†

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The Guerbet reaction of ethanol to produce heavier products was performed over a series of extensively characterized carbonate-containing hydroxyapatites (HAPs) with different Ca/P ratios and thus different densities, strengths and natures of acid and basic sites. These properties were correlated with the reactivity of the solids and an optimal ratio between the amount of acid and basic sites was evidenced (ca. 5). The best performance was accordingly obtained over the Hap-CO₃ catalyst, which gave a yield of 30% of heavier alcohols at 40% ethanol conversion.

Introduction

The Guerbet reaction, the name of which originates from Marcel Guerbet's discovery, is an organic reaction in which a primary aliphatic alcohol is converted into its β -alkylated dimer with the loss of one molecule of water. Marcel Guerbet initially found that 1-octanol, thus a primary alcohol, is transformed into C16 and C24 alcohols when heated at 200 °C in the presence of its sodium derivative or sodium derivatives of other alcohols.¹ Further, Marcel Guerbet reported that this reaction is also applicable to secondary alcohols by demonstrating the production of higher chain alcohols, namely 4-methylpentanol and 2,4-dimethylheptanol, from isopropanol with, *e.g.*, 3-methylbutyric acid, acetic acid and formic acid formed as by-products.² Weizmann *et al.* reported that the presence of an α -methylene group in the immediate vicinity of the hydroxyl group is a requisite for the aldol condensation step and hence for the Guerbet reaction. They have also confirmed that the Guerbet reaction between primary

and secondary alcohols is predetermined by the fact that an aldehyde condenses with a ketone, most of the time giving an alkylidene ketone even though there are some exceptions.^{3,4}

Bioethanol is one of the major compounds produced in large quantities from biomass. While its high vapour pressure, affinity for water and lower energy content can be considered as hurdles to an extensive application in existing fuels, its use in fuels constitutes its main output at present. Biobutanol, which is another primary alcohol that can be used not only as a fuel or a fuel additive but also as a platform molecule from which C4 chemistry can be derived, can also be produced directly from biomass, but through routes which are still not economically feasible at the moment (biotech and syngas conversion).^{5,6} Thus, finding an alternative chemical route to transform ethanol into heavier alcohols such as butanol could represent an attractive option with respect to the biorefinery concept of producing a diversity of chemicals and fuels.⁷ Such transformation is possible through the Guerbet reaction.

Both homogeneous and heterogeneous catalysts have been reported for the production of heavier alcohols through the Guerbet reaction. The Guerbet reaction using homogeneous catalysts was first examined by using transition metal complexes such as Rh, Ru, Pt, and Ir under very mild conditions of 110–140 °C at atmospheric pressure.^{8–11} The catalytic activity of the Rh complex was found to be very high compared with that of Ru and Ir complexes. Pd and Au complexes are inactive in the Guerbet reaction because it is difficult to generate stable metal hydride complexes.^{8,9} Carlini *et al.* disclosed the Guerbet reaction using homogeneous Ni- and Pd-based catalysts in the presence of basic alkoxides, but at a higher reaction temperature of 200 °C.^{9,12–14} Catalytic systems based on Cu, Ni, Rh, Ru and Pd metals in combination

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with soluble alkoxides as basic components were also employed in the Guerbet reaction to synthesize isobutanol.^{13–16} The main disadvantage of using an excess amount of alkoxides is linked to metal leaching and also to the production of alkali hydroxides like sodium hydroxide, which is not environmentally friendly.^{11–16} Thus, in order to make the Guerbet reaction economically feasible for industrial applications, there is a need for developing heterogeneous catalysts that should also be environmentally friendly in terms of catalyst separation and reusability.

As we will detail in the next sections, the Guerbet reaction needs multifunctional catalysts, especially having both acid and base sites. Ueda *et al.* reported the condensation of various primary alcohols with methanol over solid basic catalysts such as MgO, CaO and ZnO.^{17–20} The reaction took place only over MgO, as the other metal oxides were found to be inactive or non-selective. The Guerbet reaction of alcohols has also been studied using various Mg–Al mixed oxides derived from hydrotalcite-type precursors.^{21–25} The rate of alcohol dehydration to ethers and olefins is higher for Al-rich Mg_yAlO_x solids, but dehydrogenation and aldol condensation of ethanol to the corresponding aldehyde and higher alcohols, respectively, are favoured over Mg-rich Mg_yAlO_x samples.^{21,25} Also, introduction of copper into the hydrotalcite-derived Mg–Al mixed oxide enhanced the activity of the catalyst towards the Guerbet reaction.^{22,23}

Hydroxyapatite was especially found to be an active catalyst for the heterogeneous gas phase reaction of ethanol.^{26,27} In this respect, Tsuchida *et al.* reported the ethanol conversion over hydroxyapatites with different Ca/P ratios up to the stoichiometric value of 1.67.²⁷ As a result, they obtained about 70% of selectivity to higher alcohols, at an ethanol conversion of 50%. Indeed, a drawback of using hydroxyapatites for the Guerbet reaction is that at higher ethanol conversions, the produced primary alcohols undergo further condensation to produce heavier alcohols and other heavier products.²⁸ These heavier alcohols can however be used, *e.g.*, as solvents, fuel additives or plasticizers, provided that relevant separation operations are performed.

Concerning the reaction mechanism, it is most largely confirmed that the Guerbet reaction proceeds through a series of consecutive steps over multifunctional catalysts (acid, base and/or redox sites), which were especially considered to explain the reactivity of ethanol over the aforementioned HAPs.^{27,29} First, dehydrogenation of the alcohol takes place over redox/acid–base sites to form aldehydes, which further undergo aldol condensation over basic sites to produce the corresponding aldol. The as-obtained aldol then undergoes dehydrogenation over acid sites to form α and β unsaturated aldehydes or ketones that further undergo hydrogenation over redox/acid–base sites to form higher alcohols. However, Scalbert *et al.*³⁰ have recently shown that upon using temperatures over 350 °C and a commercial HAP, of which the Ca/P ratio was not mentioned but is supposedly stoichiometric, another mechanism takes place, involving direct condensation of ethanol. Note that the authors have

mentioned that their thermodynamic findings may not be applicable at lower temperatures, *i.e.*, below 350 °C.

In this context, the overall objective of the present work was to achieve a fine understanding of the acid–base functionalities of the HAPs involved in the gas phase Guerbet reaction of ethanol. In this respect, we used a previously synthesised and fully characterized series of HAPs (notably with different Ca/P ratios higher than the stoichiometric value of 1.67).³¹ Since there are only few reports correlating the structural, textural and acid–base properties of hydroxyapatites (Ca/P > 1.67) with Guerbet reactivity, we specifically tracked and discussed the correlations between the observed catalytic performance and the physico-chemical properties. This enabled drawing a complete picture of the specific action of the various types of acid/basic sites present at the surface of the solids on the formation of the various Guerbet reaction products.

Experimental

Catalysts

The synthesis method for all the solids used in this study is reported elsewhere.³¹ The solids are further listed in Table 1 together with their main characteristics.

Characterisation

Techniques for extensive physico-chemical characterization of the solids, namely, X-ray diffraction (XRD), thermogravimetric analysis (TGA), scanning electron microscopy (SEM), infrared spectroscopy (FTIR), X-ray photoelectron spectroscopy (XPS), low-energy ion scattering spectroscopy (LEIS), heteronuclear magnetic resonance spectroscopy (2D NMR), temperature programmed desorption (TPD-NH₃ & TPD-CO₂), XPS after adsorption of 2-phenylethylamine (PEA-XPS) and pulse liquid chromatographic adsorption using benzoic acid, are reported elsewhere.³¹ The present paper is thus focused on the reactivity of the solids in the Guerbet reaction, in relation to the aforementioned characterization results.

The Guerbet reaction of ethanol was carried out using 0.20 g (GHSV = 15 000 mL h^{−1} g^{−1}) or 0.60 g (GHSV = 5000 mL h^{−1} g^{−1}) of a sample under the form of a powder with a particle size between 80 and 100 mesh. 20 vol.% ethanol in He gas was passed through a fixed bed glass reactor at a total

Table 1 List of the catalytic solids used in the present study and their basic characteristics (from data in ref. 31)

Type of solid	Name	Ca/P atomic ratio (ICP)	Na (wt.%)	Specific surface area (m ² g ^{−1})
Deficient	HapD	1.62	—	124
Stoichiometric	Hap	1.69	—	114
Carbonated	Hap-CO ₃	1.70	—	107
Carbonated	HapNa-CO ₃	1.72	0.64	109
Carbonate-rich	HapE-CO ₃	1.90	—	76
Carbonate-rich	HapE-Na-CO ₃	2.39	<0.3	72

flow rate of 50 mL min⁻¹. The feed and the outlet products were analyzed on-line by a gas chromatograph equipped with a Zebron ZB-Bioethanol capillary column ($l = 30$ m, $id = 0.25$ mm) connected to a FID and coupled with a mass spectrometer (MS Agilent 5975C).

The ethanol conversion, the selectivity to each product, and the corresponding yields were calculated as follows:

- Ethanol conversion (%) = $100 \times [(\text{moles of ethanol before reaction} - \text{moles of ethanol after reaction}) / \text{moles of ethanol before reaction}]$;

- Product yield (%) = $100 \times (\text{moles of C converted into the considered product} / \text{moles of C in ethanol before reaction})$;

- Product selectivity (%) = $100 \times (\text{product yield} / \text{ethanol conversion})$.

Results and discussion

Guerbet reaction products observed at iso-conversion

The first series of experiments aimed at studying the intrinsic catalytic properties of the solids by working at iso-conversion. Table 2 shows the selectivity to the main products of the Guerbet reaction of ethanol at an iso-conversion of 14%. This iso-conversion of ethanol was achieved by appropriately adjusting the temperature between 300 °C and 400 °C at GHSVs of 5000 mL h⁻¹ g⁻¹ and 15 000 mL h⁻¹ g⁻¹. As aforementioned, the direct ethanol coupling pathway proposed by Scalbert *et al.* may not be applicable at lower temperatures, *i.e.*, below 350 °C.³⁰ The iso-conversion of 14% was observed at a temperature below 350 °C for three HAPs (HapD, Hap, Hap-CO₃; Table 2). The other 3 HAPs are carbonate-rich samples, thus with properties different from those obtained by Scalbert *et al.*, who used a commercial solid most probably with a stoichiometric composition. Moreover, Scalbert *et al.* did not consider the mechanism involved in the formation of products like ethylene and ether, which are formed by dehydration of ethanol over acid sites. In the present study, *i.e.*, under the conditions and over the specific solids we used, we will see in the following sections that the observed products are in line with the sequential mechanism proposed

elsewhere,^{27,29} which then actually applies here, even if we cannot completely rule out a potential contribution of the direct pathway to the distribution of the observed products.

Butanol was the major product over all the catalysts, its selectivity being higher over the carbonated apatites, namely Hap-CO₃ (71%) and HapNa-CO₃ (59%). Formation of ethylene, which is attributed to intramolecular dehydration of ethanol, was observed in rather small quantities over all the apatites and especially over HapD (3.1%) and Hap (2%), its selectivity being less than 1.6% over the other solids. Acetaldehyde, a dehydrogenation product of ethanol, was produced over all the catalysts and especially observed as a final product over the carbonate-rich apatites, namely HapE-CO₃ (15.5%) and HapE-Na-CO₃ (19%). Formation of 2-buten-1-ol, which is produced by dehydrogenation of ethanol followed by aldol condensation and hydrogenation of the C=O bond while leaving the C=C bond non-hydrogenated, was also observed in increasing quantities when the solids were richer in carbonates. In contrast, only trace amounts of butyraldehyde were observed, which is a compound produced in a way similar to that for 2-buten-1-ol but by hydrogenation of the C=C bond while leaving the C=O bond non-hydrogenated. Small amounts of diethyl ether, resulting from intermolecular dehydration of ethanol over acid sites, as well as dienes, were also observed over all the catalysts. Further, 'other' compounds were also detected, namely higher aldehydes, higher alkenes and trace amounts of aromatics, which are formed by multiple consecutive steps (including Guerbet intermediates), by uncontrolled reaction over the various sites present at the surface of the solids.

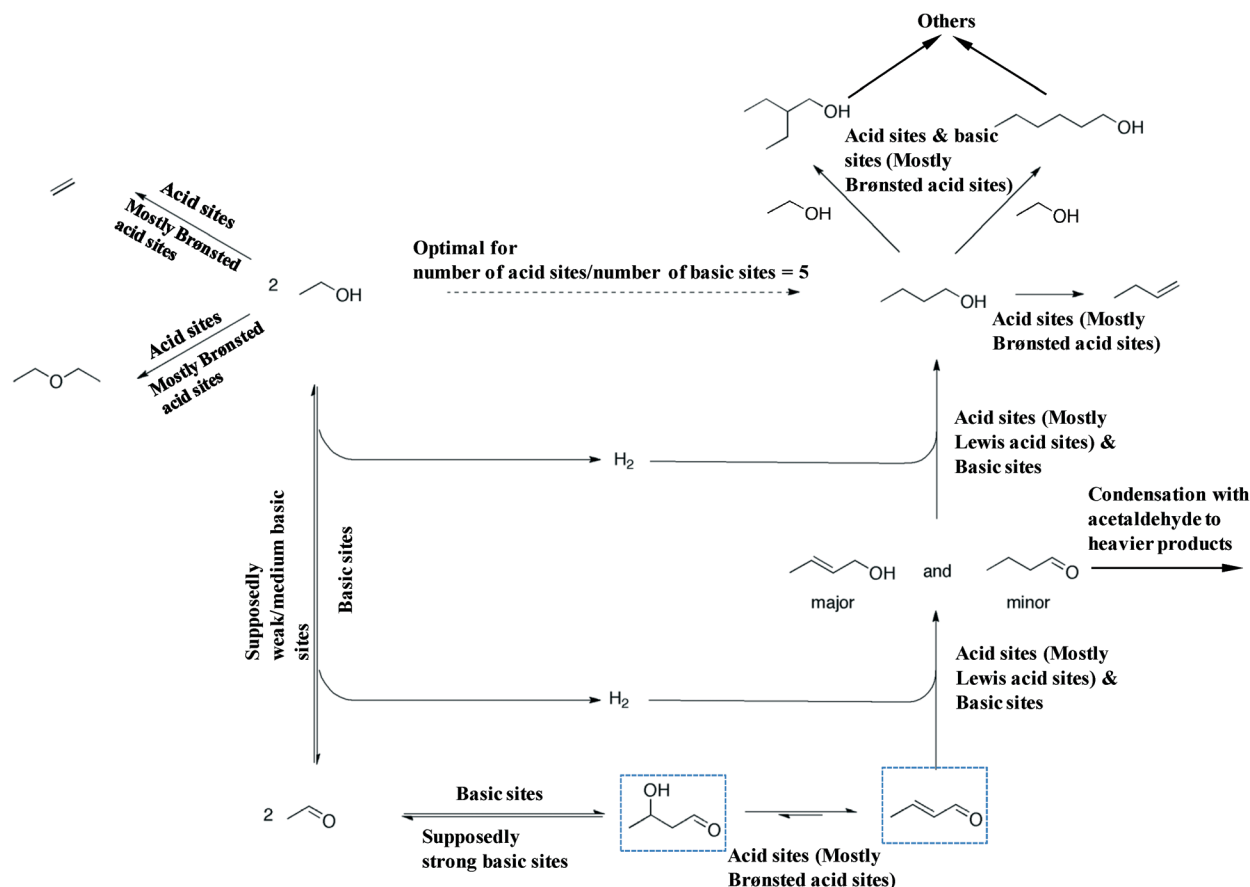
Due to the high level of complexity of the studied reaction, for the sake of clarity, we compile the conclusions deduced from all the results of this study in Scheme 1, which is thus, in fact, a conclusion of this work. Hereafter, we comment on Scheme 1 before, in the next part of the paper, detailing the specific set of correlations/conclusions that were gathered to assemble this integrated view of the whole reaction network.

In order for the Guerbet reaction of ethanol to effectively produce heavier alcohols, both acid and basic sites are

Table 2 Selectivity to Guerbet reaction products at 14% ethanol conversion

Catalysts	Group i		Group ii		Group iii	
	HapD	Hap	Hap-CO ₃	HapNa-CO ₃	HapE-CO ₃	HapE-Na-CO ₃
Ethylene	3.1	2	1.3	1.6	1.2	1.5
Acetaldehyde	5.8	9	7.2	13.1	15.5	19
2-Buten-1-ol	2	1	3.2	5.1	8	10.2
Butanol	49	50	71	59	46	41
C6-C10 alcohols ^a	18	19.4	8.8	2.9	7	1.1
Total alcohols ^b	69	70.4	83	67	61	52.3
Diethyl ether	5	4.5	1.1	0.8	0.6	2
Dienes ^c	3.1	1.2	2	2.2	1.5	1.2
Ethyl butyrate	—	—	—	1.5	3.5	10.3
Others ^d	14	12.9	5.4	13.8	16.7	13.7

^a Hexanol, 2-ethylbutanol, octanol, 2-ethylhexanol and decanol. ^b 2-Buten-1-ol, butanol and C6-C10 alcohols. ^c 1,3-Butadiene and small amounts of C6-dienes. ^d Higher aldehydes, higher alkenes and aromatics. *N.B.* GHSV = 5000 mL h⁻¹ g⁻¹ for all HAPs but 15 000 mL h⁻¹ g⁻¹ for HapNa-CO₃.



Scheme 1 Summary of the role of acid and basic sites present in the different HAPs towards different pathways. The dashed box represents the intermediates which were not observed in the reaction. [Basic sites: PO_4^{3-} , OH^- & CaO ; Brønsted acid sites: HPO_4^{2-} ; Lewis acid sites: Ca^{2+} & OH^- vacancies (δ^+)].

necessary, but the strength, nature and relative balance of these sites have great influence on each step of the Guerbet cycle, and are also involved in the occurrence of undesirable side reactions, thereby controlling the product distribution. The first step of the Guerbet cycle is the dehydrogenation of ethanol over basic sites to produce acetaldehyde. As it was concluded in our previous study on the same solids concerning the conversion reaction of isopropanol to acetone that this most probably takes place over weak or medium basic sites (CaO species),³¹ we also postulated that it might be the same here for ethanol reactivity, even if we cannot formally exclude the possibility that strong basic sites (OH^-) could also be involved. Scalbert *et al.* proposed a mechanism in which acetaldehyde can further react with unreacted ethanol to form butenol supposedly over stoichiometric HAPs at temperatures below 350 °C.³⁰ However, we used temperatures over 350 °C for the carbonate-containing apatites, which have different properties compared to stoichiometric apatites.

The as-formed acetaldehyde further undergoes aldol condensation over basic sites (probably over stronger basic sites, OH^-) to form an intermediate aldol (never observed under our conditions). Further dehydration of aldol takes place over acid sites (mostly Brønsted acid sites, HPO_4^{2-}) to produce crotonaldehyde, which was also not observed (or in trace

amounts). Crotonaldehyde then undergoes partial hydrogenation through a proton exchange mechanism over HAP catalysts to form 2-buten-1-ol and butyraldehyde over both types of acid sites (mostly Lewis acid sites, Ca^{2+} & OH^- vacancies) and over basic sites. These compounds can further undergo hydrogenation over the same acid–base sites in the apatite catalysts to form butanol. Butanol can then further react with ethanol and go through an additional Guerbet cycle involving acid and basic sites to form 2-ethylbutanol or hexanol, depending on the type of attack. The formation of these heavier alcohols was more promoted over catalysts with a larger number of acid sites, and mainly Brønsted acid sites (HPO_4^{2-}). Also, there are some competing reactions, especially dehydration reactions that can form olefins (ethylene, butene, *etc.*) or diethyl ether, which occur in parallel to the Guerbet cycle (mostly over Brønsted acid sites, HPO_4^{2-}). Thus, there should be a compromise between the number of acid and basic sites and also with regard to their relative strength and nature to yield a catalyst that is selective towards higher alcohols. We especially determined the best ratio between the number of acid and basic sites of *ca.* 5 to reach the maximum selectivity to higher alcohols over the HAP catalysts included in the present study. If the amount or the strength of the acid sites becomes too high, the

dehydration pathway (producing olefins) becomes more prominent, and if the amount or the strength of the basic sites becomes too high, this may preferably lead to formation of products like aldehydes. In contrast, if there is any deficiency in acid or basic sites, the reaction cannot proceed further towards formation of butanol or heavier alcohols, thus resulting in increased selectivity to some intermediates of the Guerbet cycle.

All these observations are derived from/supported by the conclusions of the following study, which is divided into two main parts, namely, the correlation studies between the isopropanol test³¹ and the Guerbet reaction and the direct correlation study between the Guerbet reaction and the acid-base properties of the HAPs.

Correlation studies between the isopropanol test and the Guerbet reaction. A correlation study between the selectivity to products observed when targeting the Guerbet reaction starting from ethanol (ethylene, diethyl ether, acetaldehyde, alcohols, *etc.*) and the selectivity to isopropanol reaction products (both taken at iso-conversion; see ref. 31) was performed to get an indirect insight into the acid-base behavior of all the apatite catalysts. Indeed, the isopropanol test on the apatite catalysts mainly produced propylene (together with small quantities of diisopropyl ether) and acetone, which is an indirect representation of the acidic behavior and the basic character, respectively, of the probed solids.³¹ In the following figures, the catalysts are divided into three groups (corresponding to the groups given in Table 2): (i) HapD and Hap represented by circles, (ii) carbonated apatite catalysts, namely Hap-CO₃ and HapNa-CO₃, represented by squares, and (iii) carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃, represented by triangles (this representation is similar to that in ref. 31).

Since ethylene and diethyl ether (reaction products of ethanol) and propylene and diisopropyl ether (reaction products of isopropanol) are supposed to be formed by inter- and intra-molecular dehydration reactions over acid sites, a correlation study between these products was performed. As

diisopropyl ether was hardly detected over group ii and group iii catalysts due to the GC analysis detection limit of *ca.* 0.5%, its quantity was then arbitrarily accounted as $0.25\% \pm 0.25\%$ over these samples.

Fig. 1(a) reveals a good correlation between ethylene and propylene selectivities. The selectivity to ethylene did not vary much for group ii & iii catalysts, with a value of $1.4 \pm 0.2\%$, but increased for group i catalysts, namely Hap (2%) and HapD (3.1%). HapD and Hap exhibited higher selectivity to both ethylene and propylene, thus showing their predominant acidic nature compared to the other solids. A consistent correlation was obtained for the plot of diethyl ether selectivity as a function of diisopropyl ether selectivity [Fig. 1(b)], with the presence of low amounts of both compounds over group ii & group iii catalysts and more significant amounts over group i catalysts. Thus, it can be concluded that the predominant acidic nature of group i catalysts was responsible for both inter- and intra-molecular dehydration of the respective substrates.

Fig. 2 shows the variation in acetaldehyde selectivity (dehydrogenation product of ethanol in the Guerbet reaction) with acetone selectivity (dehydrogenation product of isopropanol), both originating from a reaction over basic sites. A rather good correlation was obtained. The selectivity to acetaldehyde increased from 5.8% in HapD (group i) to 19% in HapE-Na-CO₃ (group iii) along with an increase in the acetone selectivity from 5% to 72.8%, respectively. The higher selectivity to both acetone and acetaldehyde over carbonate-rich apatites (group iii) indicated their predominant basic character compared to the other apatites. In addition, a linear increase (represented by the dashed line in Fig. 2) was observed especially for the apatites that are highly selective to acetone and acetaldehyde.

However, a direct correlation considering acetaldehyde selectivity is difficult because, at least in the considered cascade mechanism, acetaldehyde is an intermediate in the Guerbet reaction of ethanol that can be further consumed by condensation reaction to heavier products through the whole

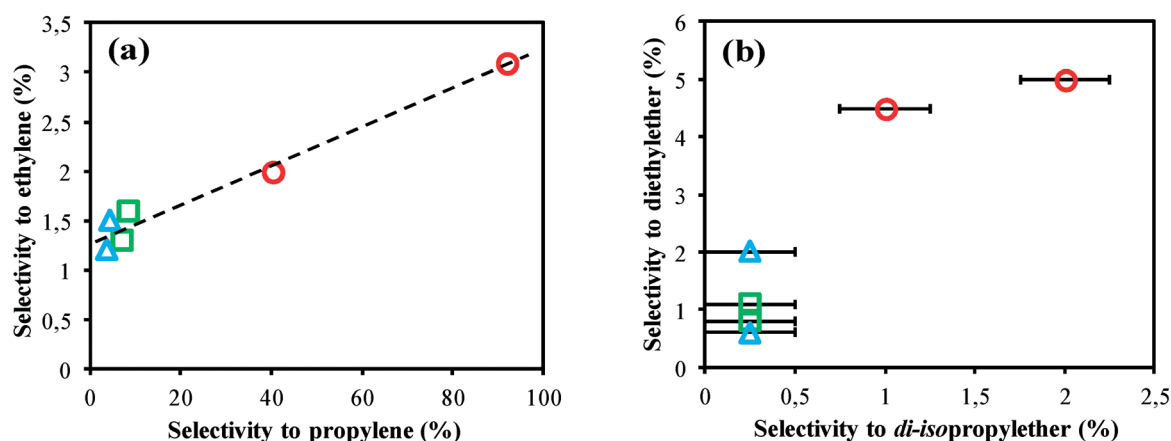


Fig. 1 (a) Selectivity to ethylene as a function of selectivity to propylene and (b) selectivity to diethyl ether as a function of selectivity to diisopropyl ether (represented by circles for HapD and Hap, squares for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and triangles for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

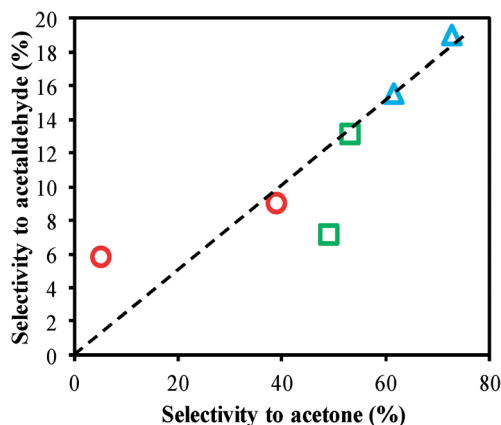


Fig. 2 Selectivity to acetaldehyde as a function of selectivity to acetone (represented by circles for HapD and Hap, squares for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and triangles for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

sequence of Guerbet reaction steps. This is not the case for acetone formation, as this product is a 'dead-end' product in the reaction of isopropanol. Therefore, there is a large chance to underestimate the basic nature of a catalyst in the ethanol conversion reaction by only considering its acetaldehyde selectivity. This may be the reason why some catalysts (e.g., Hap-CO₃, represented by the lower green square in Fig. 2) show deviation from the linearity to lower acetaldehyde selectivity, while supposedly having significant dehydrogenation properties.

As a conclusion of this part (including further relevant correlations presented in the ESI† file, namely Fig. S1, S2 & S3), interesting correlations between the Guerbet ethanol reaction and the isopropanol test were obtained. The selectivities to propylene and diisopropyl ether were both found to be correlated with the ethylene and diethyl ether selectivities, respectively, and their formation confirms the stronger acidic behavior of Hap and HapD (group i) compared to the other catalysts. The deviation observed for the highly efficient Guerbet catalyst Hap-CO₃ further confirmed the difficulty in the direct correlation of selectivity to acetaldehyde, which is an intermediate in the Guerbet reaction. Selectivity to C6–C10 alcohols were larger over group i catalysts, which are more acidic by nature, thus revealing the fact that the presence of acid sites facilitates butanol and ethanol condensation. Almost all the observations clearly indicated that very fine tuning of the apatite catalysts is necessary to avoid competitive side reactions like dehydration and have a maximum selectivity to Guerbet alcohols.

Direct correlation studies between the Guerbet reaction and acid–base properties of the apatites. After indirect correlation with the results of the isopropanol test, direct correlation of the Guerbet reactivity with the measured acid–base properties³¹ of the solids was studied.

Fig. S4† shows the selectivity to target products (butanol or mix of higher alcohols) on plots reporting acidity as a function of basicity of the catalysts (in terms of amount or

density). It highlights that Hap-CO₃ is not so far from being an ideal catalyst that possesses well-balanced acid–base properties, enabling a maximum selectivity to butanol or total alcohols. We plotted a dashed line from the origin to the point representing Hap-CO₃. On this line, the relative density of basic to acid sites was measured to be 0.2 (slope). Then, we can suppose that any apatite catalyst (or if we may generalize this assumption, that any solid) lying on this line shows a good compromise between acid and basic sites and can give a maximum selectivity to butanol or total alcohols. Hence, the ideal HAP catalyst should have a number of acid sites 5 times greater than that of basic sites (furthermore, should be well distributed on the surface to enable the Guerbet reaction sequence, thus using antagonist sites that are supposedly close to one another). The catalysts that are in the region above this line possess too many basic sites. This is the reason why HapNa-CO₃ shows a lower selectivity to butanol or total alcohols though it possesses a higher amount of basic sites compared to the so-called 'ideal' Hap-CO₃ catalyst. This is the same for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃, with basicity/acidity ratios (slopes) of 0.4 and 0.7, respectively, compared to the 'ideal' value of 0.2. Moreover, the catalysts that lie below the dashed line possess a very large proportion of acid sites compared to the ideal Hap-CO₃ catalyst. This is the reason why Hap and HapD with basicity/acidity ratios of 0.04 and 0.13, respectively, show lower butanol or total alcohol selectivities compared to Hap-CO₃, though they possess higher amounts or densities of acid sites than Hap-CO₃. After this first observation, in the following sections, we systematically looked for potential direct correlations between the number/nature of sites and the selectivity to the main Guerbet products. This enabled revealing global trends, corroborating the above observations in Fig. S4.†

Selectivity to ethylene and diethyl ether as a function of acid–base properties of the HAP catalysts. Selectivity to ethylene remained almost unchanged, though the number of acid sites increased from 141.3 μmol g^{−1} (HapE-Na-CO₃) to 658 μmol g^{−1} (Hap-CO₃), as shown by the dashed line parallel to the x-axis in Fig. 3. In contrast, a further increase in the amount of acid sites to 763.3 μmol g^{−1} (Hap) and 956 μmol g^{−1} (HapD) yielded a linear increase in ethylene selectivity to 2% and 3.1%, respectively. This, again, as shown using the comparison with the isopropanol reactivity test, clearly indicates the predominant acidic behavior of Hap, and especially of HapD, compared to the other samples.

We then deepened the interpretation of the results by taking into account the nature of the acid sites (previously determined in† using PEA-XPS), and the results suggested that ethanol dehydration to ethylene most likely occurs on Brønsted acid sites (HPO₄^{2−}) in HAPs (see Fig. S6 and associated comments in the ESI† file).

The same kind of conclusion was drawn for diethyl ether formation, which also seemed to strongly depend on the presence of Brønsted acid sites in HAPs (see Fig. S7 & S8 and associated comments in the ESI† file).

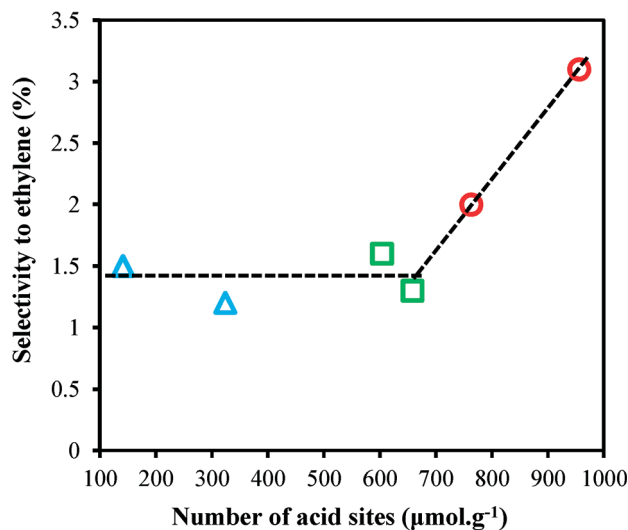


Fig. 3 Selectivity to ethylene as a function of the number of basic sites (represented by circles for HapD and Hap, squares for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and triangles for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

Then, the inter- and intra-molecular dehydration reactions of ethanol mainly proceed over Brønsted acid sites (HPO_4^{2-}) in HAPs. In this respect, it is also worth mentioning that the Guerbet reaction produces water. Thus, at least some Lewis sites observed during *ex situ* measurement might undergo hydration and be converted to Brønsted sites, which can either be used for catalysis or be dehydrated to recover the initial Lewis sites under the reaction conditions, during the reaction. Equilibrium should be reached, but it is very difficult to predict the proportion between species at the steady state.

Selectivity to acetaldehyde as a function of acid–base properties of the HAP catalysts. Fig. 4 shows the variation in

acetaldehyde selectivity with the number of acid and basic sites. It might appear difficult to draw a correlation between acetaldehyde selectivity and acidic or basic properties. Indeed, while this reaction occurs on basic sites, as aforementioned, acetaldehyde is an intermediate in the Guerbet reaction of ethanol and should then be further consumed during the reaction over efficient catalysts. This explains why a linear correlation could not be observed between acetaldehyde selectivity and the number of basic sites [Fig. 4(a)]. In contrast, a very interesting linear correlation between acetaldehyde selectivity and the number of acid sites was observed [Fig. 4(b)]. The acetaldehyde selectivity was higher over carbonate-rich apatites HapE-Na-CO₃ (19%) and HapE-CO₃ (15.5%) with lower amounts of acid sites (141.3 $\mu\text{mol g}^{-1}$ and 324 $\mu\text{mol g}^{-1}$, respectively) compared to those over the other solids. We suppose that the carbonate-rich apatites lack sufficient strong basic sites (OH^- groups),³¹ which assist in the condensation of acetaldehyde further to heavier Guerbet products (see Scheme 1). The acetaldehyde selectivity decreased to 13.1% (HapNa-CO₃) and 7.2% (Hap-CO₃) with an increase in the number of acid sites to 604 $\mu\text{mol g}^{-1}$ and 658 $\mu\text{mol g}^{-1}$, respectively. The lower acetaldehyde selectivity over Hap and HapD is not only due to condensation of acetaldehyde, but also because of an increased formation of ethylene and diethyl ether, in competition with the formation of the intermediate acetaldehyde over basic sites, which are two reactions primarily consuming ethanol. Hence, the amount of acid sites can indirectly influence the apparent acetaldehyde selectivity in two ways.

Selectivity to 2-buten-1-ol as a function of acid–base properties of the HAP catalysts. Since 2-buten-1-ol is an intermediate in the Guerbet reaction of ethanol to butanol, finding a direct correlation of its selectivity with the acid–base properties of the solids is rather difficult as in the case of acetaldehyde. However, to identify whether potential indirect

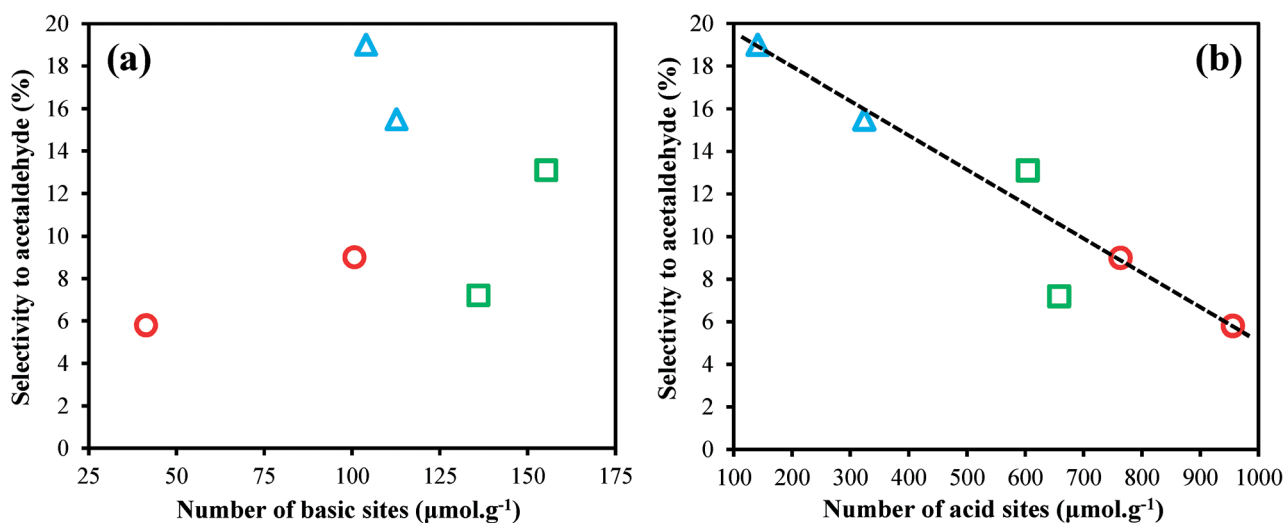


Fig. 4 Selectivity to acetaldehyde as a function of (a) the number of acid sites and (b) the number of basic sites in HAPs (represented by circles for HapD and Hap, squares for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and triangles for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

correlations can arise, a systematic correlation study was also performed for this compound (Fig. S9 & S10 and related comments in the ESI†). This study shows that the Lewis acid sites (Ca^{2+} & OH^- vacancies) in HAPs are responsible for the transformation of 2-buten-1-ol to butanol (in contrast with the Brønsted sites, which are responsible for dehydration reactions).

Selectivity to Guerbet alcohols as a function of acid–base properties of the HAP catalysts. It is difficult to find a direct correlation between Guerbet alcohol selectivities (butanol, saturated alcohols or total alcohols) and the number of acid sites. However, a rather linear increase (represented by the dashed line for total alcohols) in all the Guerbet alcohol selectivities was observed with an increase in the number of acid sites from $141.3 \mu\text{mol g}^{-1}$ (HapE- Na-CO_3) to $658 \mu\text{mol g}^{-1}$ (Hap- CO_3), as shown in Fig. 5(a).

Hap and HapD again showed deviation from linearity due to their higher number of acid sites ($763.3 \mu\text{mol g}^{-1}$ and $956 \mu\text{mol g}^{-1}$, respectively). Selectivity to butanol, saturated

alcohols or total alcohols remained unchanged even when the number of acid sites increased from $763.3 \mu\text{mol g}^{-1}$ (Hap) to $956 \mu\text{mol g}^{-1}$ (HapD), which shows that too much increase in the number of acid sites has little influence on Guerbet alcohol selectivity. The difference between the grey-filled symbols (saturated alcohol selectivity) and open symbols (butanol selectivity) represents the formation of C6–C10 alcohols, and this difference is higher over Hap and HapD compared to that over the other catalysts. It is an indirect representation showing the ability of these apatites with predominant acidic behavior to further condense butanol & ethanol to C6–C10 alcohols. All these observations are in good agreement with Fig. S1(a)† showing the relationship between Guerbet alcohol selectivity and propylene selectivity. In contrast, a consistent correlation between butanol, saturated alcohols or total alcohols and the number of basic sites was not obtained, as shown in Fig. 5(b). However, a difference between the grey-filled symbols and open symbols, which represents the

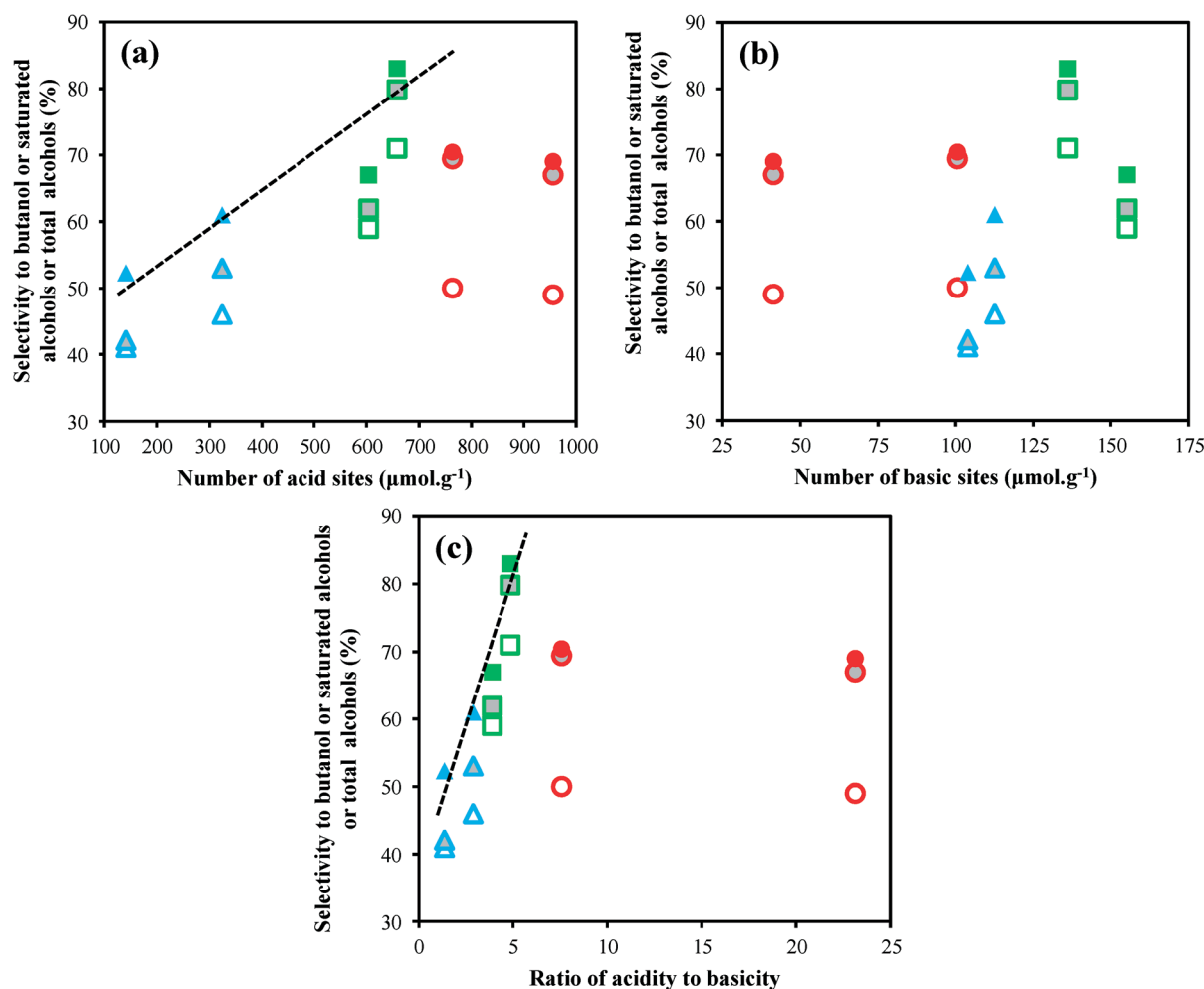


Fig. 5 (a) Selectivity to butanol, saturated alcohols or total alcohols as a function of the number of acid sites, (b) selectivity to butanol, saturated alcohols or total alcohols as a function of the number of basic sites, and (c) selectivity to butanol, saturated alcohols or total alcohols as a function of the ratio of the number of acid sites to the number of basic sites. Selectivity to butanol is represented by open symbols, while selectivity to saturated alcohols (excluding 2-buten-1-ol) and heavier alcohols is represented by grey-filled symbols and color-filled symbols, respectively (red circles for HapD and Hap, green squares for carbonated apatites, namely Hap- CO_3 and HapNa- CO_3 , and blue triangles for carbonate-rich apatites, namely HapE- CO_3 and HapE- Na-CO_3).

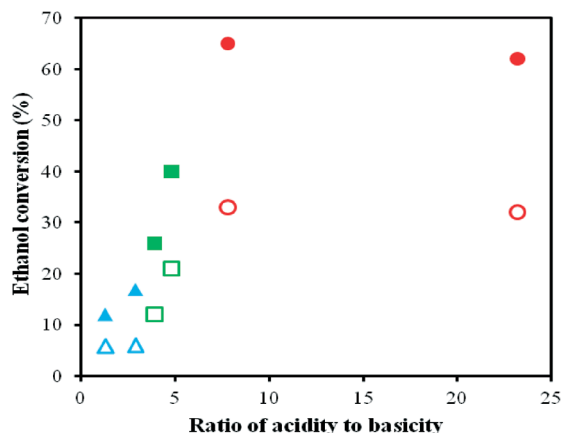


Fig. 6 Ethanol conversion as a function of the ratio of acidity to basicity at 400 °C; GHSV = 5000 mL h⁻¹ g⁻¹ (full symbols); GHSV = 15000 mL h⁻¹ g⁻¹ (open symbols) (represented by circles for HapD and Hap, squares for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and triangles for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

formation of C6–C10 alcohols, is clearly observed, although no consistent trend could be revealed.

Since the production of Guerbet alcohols is subject to the contribution of both acid and basic sites, the Guerbet alcohol (butanol, saturated alcohols or total alcohols) selectivity as a function of the ratio of the number of acid sites to that of basic sites was plotted [Fig. 5(c)]. The trend was similar to that observed when plotting selectivity as a function of the number of acid sites, as in Fig. 5(a). The selectivity to all the Guerbet alcohols sharply increased from carbonate-rich apatite HapE-Na-CO₃ to carbonated apatite Hap-CO₃ with an increase in the acidity-to-basicity ratio from 1.3 to 4.8. A further increase in the acidity-to-basicity ratio from 4.8 (Hap-CO₃) to 7.6 (Hap) led to a decrease in Guerbet alcohol

selectivity. The selectivity to butanol, saturated alcohols or total alcohols remained unchanged upon an increase in the acidity-to-basicity ratio to a maximum of 23.1. The above observations are in good agreement with the variation in Guerbet alcohol selectivity and the propylene/acetone ratio, which is an indirect representation of the acid/base ratio (Fig. S2†).

Guerbet reaction: optimization of experimental conditions

In the literature, we can find studies related to the Guerbet reaction of ethanol over hydroxyapatites, but only few studies which focus on maximizing the selectivity to butanol at higher ethanol conversions were reported.²⁷ Although the authors in ref. 27 reported a similar selectivity to higher alcohols at 50% ethanol conversion, it is difficult to compare with our study since we used different GHSVs and temperatures. The aim of optimization in the present work was to enhance the ethanol conversion so as to increase the yield of heavier alcohols, instead of aiming at only synthesizing butanol. This helped us to draw general behavior lines concerning the Guerbet reactivity and the acid–base properties of the solids. In this part, the ethanol reaction was performed at different temperatures and GHSVs in order to approach optimal conditions for the production of heavier alcohols.

As a general summary of the observed results (Fig. 6 and Tables 3 & 4), before detailing specific points in the following sections, it appears that the reaction products consisted of alkenes, dienes, aldehydes, alcohols, diethyl ether, ethyl butyrate and aromatics, the major products being ethylene, acetaldehyde, and C4–C10 alcohols including significant amounts of butanol. Condensation products with only even numbers of carbon atoms like C4, C6, C8, *etc.* were observed, indicating that the products formed during the reaction were not subjected to any mechanism involving C–C bond cleavage. Only trace amounts of light gaseous products such as

Table 3 Conversion and product distribution over hydroxyapatite catalysts at GHSV = 15 000 mL h⁻¹ g⁻¹

	HapD	Hap	Hap-CO ₃	HapNa-CO ₃	HapE-CO ₃	HapE-Na-CO ₃
Conversion at						
300 °C	4.20	4.50	4.0	3.0	1.0	0.50
350 °C	10.40	11.0	8.60	6.70	3.30	2.20
400 °C	32	33	21	12	6	5.80
Yield at 400 °C						
Alkenes						
Ethylene	12.8	5.6	0.44	0.19	0.09	0.06
Butene	0.80	1.10	—	—	—	—
Hexene	0.10	0.30	0.04	—	—	—
Aldehydes						
Acetaldehyde	2.50	2.10	1.20	1.60	1.30	1.40
Butanal	0.10	0.26	0.15	0.14	0.13	—
Dienes						
Butadiene	3.30	2.70	0.50	0.30	0.10	—
C6-dienes	0.48	1.20	0.11	—	—	—
Alcohols						
Butanol	7.80	14.90	12.60	7.0	3.60	3.0
C6–C10 alcohols	0.70	2.50	2.20	1.0	0.2	—
Ethyl butyrate	—	—	0.08	0.18	0.39	0.81
Aromatics	—	—	0.19	0.14	0.12	0.09

Table 4 Conversion and product distribution over hydroxyapatite catalysts at GHSV = 5000 mL h⁻¹ g⁻¹

	HapD	Hap	Hap-CO ₃	HapNa-CO ₃	HapE-CO ₃	HapE-Na-CO ₃
Conversion at						
300 °C	14.2	13	5.8	5.4	4.8	4.5
350 °C	27	32.2	15.2	9.5	9	6.8
400 °C	62.1	65.1	40	26	17	12
Yield at 300 °C						
Alkenes						
Ethylene	0.4	0.1	0.1	0.02	0.1	0.2
Aldehydes						
Acetaldehyde	0.8	1.8	0.3	0.5	1.0	1.8
Butanal	—	—	—	—	0.01	—
Dienes						
Butadiene	0.2	0.2	0.04	0.02	—	—
C6-dienes	0.3	—	—	—	—	—
Alcohols						
Butanol	7.1	6.5	4.2	3.7	2.2	1.1
C6–C10 alcohols	3.2	2.6	0.5	0.4	0.4	—
2-Buten-1-ol	0.1	0.1	0.2	0.4	0.4	0.5
Diethyl ether	0.7	0.6	0.04	0.03	—	—
Ethyl butyrate	—	—	—	—	0.1	0.5
Yield at 350 °C						
Alkenes						
Ethylene	2.6	1.4	0.2	0.1	0.1	0.2
Butene	0.3	0.5	—	—	—	—
Hexene	0.1	0.2	—	—	—	—
Aldehydes						
Acetaldehyde	0.8	2.6	0.7	1.1	1.5	2.0
Butanal	—	0.3	0.1	—	0.1	—
Dienes						
Butadiene	1.4	1.2	0.3	0.1	0.05	0.03
C6-dienes	0.7	0.6	0.1	—	—	—
Alcohols						
Butanol	15.1	18.2	10.6	6.2	4.4	2.9
C6–C10 alcohols	3.5	5.4	1.4	0.3	0.7	0.1
2-Buten-1-ol	0.2	0.1	0.5	0.5	0.7	0.5
Diethyl ether	1.6	1.2	0.2	0.1	0.1	0.3
Ethyl butyrate	—	—	—	0.1	0.3	0.7
Aromatics	—	—	0.1	0.3	0.3	0.04
Yield at 400 °C						
Alkenes						
Ethylene	36.7	11.4	0.6	0.4	0.2	0.3
Butene	3.0	2.8	0.1	—	—	—
Hexene	0.3	2.7	0.04	—	—	—
Aldehydes						
Acetaldehyde	1.9	3.1	1.4	1.6	2.6	2.3
Butanal	—	0.8	0.5	0.2	0.4	0.2
Dienes						
Butadiene	2.4	3.3	1.1	0.7	0.2	0.1
C6-dienes	0.7	5.3	0.3	0.1	0.1	—
Alcohols						
Butanol	6.4	21.5	22.4	15.2	7.7	4.9
C6–C10 alcohols	0.2	3.1	6.8	2.6	1.4	0.3
2-Buten-1-ol	0.1	0.2	0.8	1.2	1.2	1.1
Diethyl ether	5.0	2.0	0.2	0.2	0.1	0.2
Ethyl butyrate	—	—	0.1	0.3	0.6	1.2
Aromatics	—	—	0.9	1.0	1.1	0.8

CO₂ were observed, confirming that ethanol or the reaction products were not further decomposed over hydroxyapatite catalysts, which is in agreement with previous reports.²⁶

The ethanol conversion observed at 400 °C and at two different GHSVs we used (namely, 5000 mL h⁻¹ g⁻¹ and 15 000 mL h⁻¹ g⁻¹, obtained using 0.2 g and 0.6 g of the catalyst, respectively) is shown in Fig. 6. The ethanol conversion

increased with decreasing GHSV from 15 000 mL h⁻¹ g⁻¹ to 5000 mL h⁻¹ g⁻¹.

To generalize, ethanol conversion increased with an increase in the ratio of the number of acid sites to that of basic sites from 1.3 (HapE-Na-CO₃) to 7.8 (Hap), which then remained unchanged with further increase to 23.2 (HapD). This is consistent with the results observed at iso-conversion,

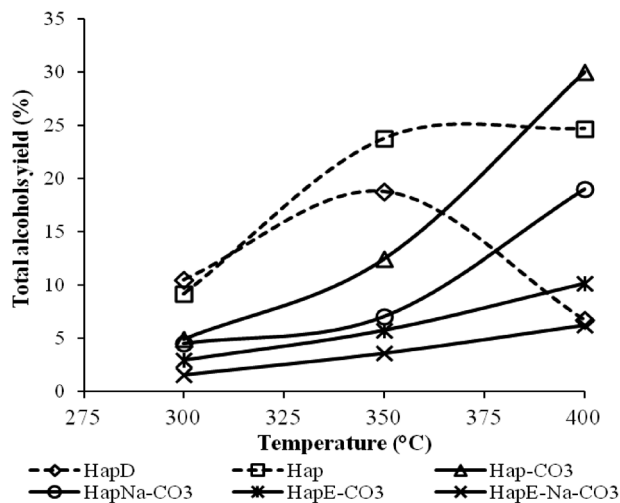


Fig. 7 Influence of temperature on heavier alcohol yield (GHSV = 5000 mL h⁻¹ g⁻¹).

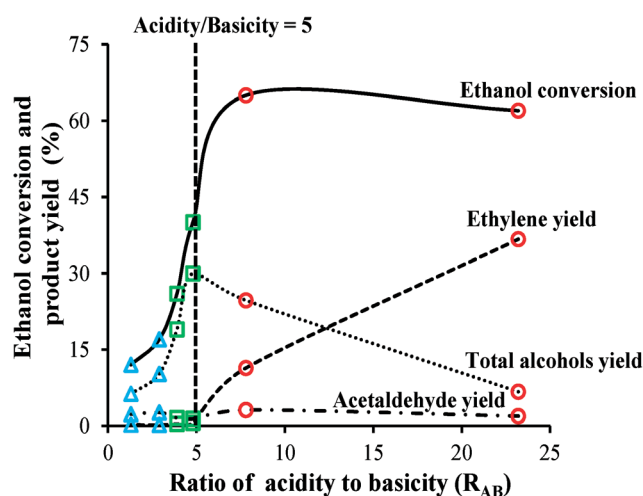


Fig. 8 Ethanol conversion and major product yields at 400 °C and GHSV = 5000 mL h⁻¹ g⁻¹ as a function of the ratio of number of acid sites to the number of basic sites (represented by red colored symbols for HapD and Hap, green colored symbols for carbonated apatites, namely Hap-CO₃ and HapNa-CO₃, and blue colored symbols for carbonate-rich apatites, namely HapE-CO₃ and HapE-Na-CO₃).

revealing that a very large increase in the amount of acid sites in the HAP catalysts had no positive effect on the ethanol conversion. We would like to point out that ethanol conversion can be modulated by the SSA of the HAPs. However, it is clear that the carbonated (Hap-CO₃ and HapNa-CO₃) and stoichiometric (Hap) HAPs having close surface areas showed entirely different conversions (Tables 1, 3 & 4), underlining the preponderant chemical effect of the acid–base sites.

The yields of minor products such as diethyl ether, ethyl butyrate and aromatics increased with a decrease in GHSV, indicating that dehydration and some dehydrogenation/polymerization reactions were promoted at lower GHSV. Selectivity to acetaldehyde decreased with a decrease in GHSV, suggesting that aldol condensation of acetaldehyde to heavier products became more prominent due to an increase in the

resulting acetaldehyde contact time (Tables 3 & 4). C4–C10 alcohol selectivity increased with decreasing GHSV for all the catalysts except for HapD and Hap, over which dehydration products like ethylene, diethyl ether and dienes, resulting from competing side reactions on acid sites, were detected. Since our aim here was to maximize ethanol conversion while yielding a maximal amount of total alcohols, further studies were performed at GHSV = 5000 mL h⁻¹ g⁻¹.

Except for Hap and HapD (represented by dashed lines), the C4–C10 total alcohol yield increased while increasing the temperature up to 400 °C (Fig. 7). Above this temperature, the yield towards aromatics and olefins severely increased, which subsequently decreased the alcohol yield (data not shown). In Hap, the total alcohol yield remained constant at 350 °C and 400 °C, whereas in HapD, the optimum temperature for the production of heavier alcohols was observed to be 350 °C. The volcano curve obtained for HapD and the plateau for Hap clearly reflect the highly acidic behavior of HapD compared with Hap. HapD and Hap have shown almost similar selectivity to Guerbet alcohols and the difference in their behaviors was not much evident during iso-conversion studies, where ethanol conversion of HapD and Hap was maintained at 14% at lower temperature (~300 °C). It is clearly observed from the dashed lines in Fig. 7 that the deviation of the total alcohol yield between HapD and Hap tended to increase with an increase in temperature. This observation clearly underlines an enhancement in the acidic behavior of HapD, which possesses a higher number of acid sites, especially Brønsted acid sites, thus leading to a higher ethylene yield compared to that of Hap (Table 4).

Further, as aforementioned, the behavior of HapD and Hap (represented by dashed lines) was different from that of the other catalysts. This discrepancy could also be related to an increase in ethylene (and diethyl ether) yield for Hap and HapD, which is not observed for the other hydroxyapatites (Table 4). This increase is much more pronounced for the HapD catalyst in comparison with that for Hap. For HapD, the ethylene yield increased from 2.6% to 36.7% and for Hap, from 1.4% to 11.4% when the temperature increased from 350 °C to 400 °C, respectively. This is in good agreement with previous results at iso-conversion, where HapD and Hap showed deviations from other catalysts due to their highly acidic behavior. As aforementioned, dehydration of ethanol to ethylene (and to diethyl ether) is the main competing reaction, which is responsible for the decrease in heavier alcohol yields. The temperature of 400 °C was found to be the most favorable temperature for the production of Guerbet alcohols over all the catalysts except for HapD and the following study was focused on this temperature.

Fig. 8 shows the variation in the ethanol conversion and yield in the major products as a function of the ratio of the number of acid sites to the number of basic sites of the hydroxyapatites at 400 °C and a GHSV of 5000 mL h⁻¹ g⁻¹. As aforementioned, ethanol conversion increased with increasing acidity/basicity ratio up to a certain limit and then remained unchanged with further increase of this ratio. The

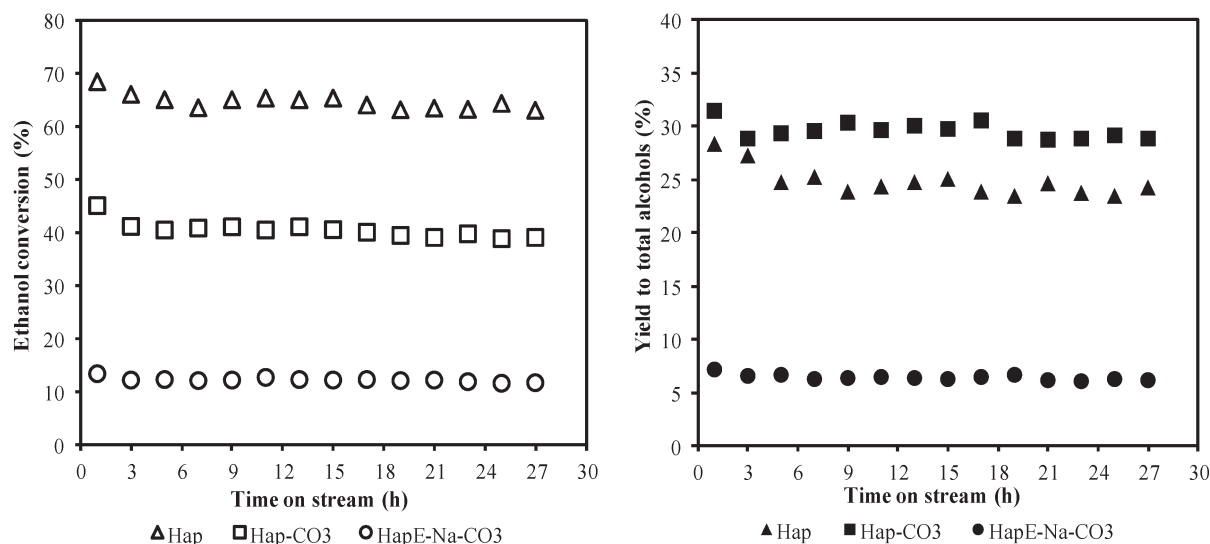


Fig. 9 Ethanol conversion and total alcohol yield over stoichiometric (Hap), carbonated (Hap-CO₃) and carbonate-rich (HapE-Na-CO₃) hydroxyapatites at 400 °C and GHSV = 5000 mL h⁻¹ g⁻¹ as a function of time-on-stream.

yield of total alcohols increased rather linearly from 6.3% to 30% with increasing acidity/basicity ratio from 1.3 (HapE-Na-CO₃) to 4.8 (Hap-CO₃), respectively, and then decreased with further increase in the acidity/basicity ratio. Hence, the Hap and HapD catalysts with higher acidity/basicity ratios of 7.8 and 23.2 showed total alcohol yields of 24.8% and 6.7%, respectively. Hap-CO₃ with an acidity/basicity ratio of 4.8 exhibited superior performance in terms of maximum total alcohol yield, which is in good agreement with the results of the iso-conversion study.

The catalysts Hap and HapD obviously deviate from linearity due to their high acidity/basicity ratio (highly acidic behavior), which is represented by higher ethylene yields of 11.4% and 36.7%, respectively. The ethylene yield was almost similar for all the other catalysts with acidity/basicity ratios of ≤ 5 . All the above observations are in good agreement with the iso-conversion studies, where Hap-CO₃ behaved as an ideal catalyst and Hap and HapD as highly acidic catalysts. It is worth mentioning that the yield of total alcohols over HapD drastically decreased compared to that obtained in the iso-conversion studies, which is explained by the dehydration of ethanol to ethylene over acid sites, and became more pronounced at a higher temperature of 400 °C. Since ethylene is a product directly formed from ethanol in the presence of acidic catalysts (primary reaction product), we suppose that it can be formed more easily at higher temperatures when catalytic reactivity is enhanced, compared to higher alcohols, which are formed from ethanol through a multistep reaction. The yield of acetaldehyde remained almost unchanged with an increase in the acidity/basicity ratio. However, the selectivity to acetaldehyde remained higher over carbonate-rich apatites HapE-CO₃ (15.5%) and HapE-Na-CO₃ (19%) compared to that over the other catalysts, as observed in the iso-conversion studies, showing the lack of a sufficient amount of acid sites to condense acetaldehyde further to aldol.

Further, the observed yield of acetaldehyde was low on the left side of the 'ideal' dashed line in Fig. 8 because it was consumed through the Guerbet reaction after formation, while its yield was low on the right side of the same line, in part due to Guerbet reaction also, but mainly due to preferential consumption of ethanol by dehydration to give ethylene (and diethyl ether).

In order to obtain insight into the effect of carbonate substitution on the stability of the catalytic performance, ethanol conversion was recorded for more than 24 h over three selected (stoichiometric, carbonate and carbonate-rich) catalysts (Fig. 9). We found that all the catalysts were quite stable, without deactivation, which suggests the potential industrial relevance of such systems.

Conclusion

As a conclusion, ethanol conversion was higher over HapD and Hap due to their acidic behaviors compared to that over the other catalysts. It was found that the acidic behavior of HapD became predominant at higher temperatures and resulted in higher ethylene yield. The ethanol conversion and total alcohol yield were found to be optimum over the Hap-CO₃ catalyst with an acidity/basicity ratio of nearly 5, which is in good agreement with the iso-conversion studies. For the catalysts that possess acidity/basicity ratios < 5 , ethanol conversion was lower, which resulted in lower yield of total alcohols. In contrast, the catalysts with acidity/basicity ratios > 5 (Hap & HapD) have shown higher ethanol conversion, but were found to be less selective to total alcohols because of an increase in ethylene selectivity, thereby decreasing total alcohol selectivity. Hence, it is confirmed that fine tuning of acid-base sites (both their nature and amount) in HAPs is necessary to have higher ethanol conversion without losing Guerbet alcohol selectivity.

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